Software Requirements Specification

Traffic Management System

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-01	Functional	User Registration: The system shall allow a new user to create an account using required personal and authentication details.	High	Pass: A new user can submit all required details and an account is created. Fail: Missing or invalid required details are accepted, or registration fails for valid details.	Enables users to access system features through individual accounts.	
FR-02	Functional	User Login: The system shall authenticate registered users and provide access according to their role.	High	Pass: Valid registered credentials authenticate the user and access matches the user's role. Fail: Invalid credentials are accepted or role access is incorrect.	Provides secure entry to the system and role-appropriate functionality.	
FR-03	Functional	Role-Based Access: The system shall restrict administrator and emergency functions to authorized users.	High	Pass: Only authorized roles can access administrator and emergency functions. Fail: An unauthorized user can access a restricted function.	Protects sensitive controls and emergency functions from unauthorized use.	
FR-04	Functional	Traffic Monitoring: The system shall display current or simulated traffic conditions and congestion levels for supported roads.	High	Pass: Traffic conditions and congestion levels are displayed for supported roads using available current or simulated data. Fail: Supported-road data is not displayed when available.	Gives users visibility into traffic conditions.	
FR-05	Functional	Live Traffic Map: The system shall display an interactive map containing roads, traffic density, congestion, and reported incidents.	High	Pass: The map is interactive and shows roads, traffic density, congestion, and reported incidents when data is available. Fail: The map is not interactive or omits required map information.	Combines traffic information in a single visual interface.	
FR-06	Functional	Signal Monitoring: The system shall allow authorized administrators to view the status of traffic signals.	Medium	Pass: An authorized administrator can view traffic-signal status. Fail: Status is unavailable to an authorized administrator or exposed through unauthorized access.	Supports monitoring of traffic signal operations.	
FR-07	Functional	Signal Control: The system shall allow authorized administrators to change signal status in the prototype, subject to validation and logging.	High	Pass: An authorized administrator can change signal status in the prototype and the action is validated and logged. Fail: An invalid change is applied or the action is not logged.	Enables controlled prototype-level signal management with traceability.	
FR-08	Functional	Incident Reporting: The system shall allow users to report accidents, roadblocks, hazards, and other traffic issues.	High	Pass: A user can submit a report for an accident, roadblock, hazard, or other traffic issue. Fail: A supported incident report cannot be submitted.	Collects traffic incident information from users.	
FR-09	Functional	Incident Management: The system shall allow administrators to review, verify, update, and close incident reports.	High	Pass: An administrator can review, verify, update, and close incident reports. Fail: Any required management action is unavailable to an authorized administrator.	Allows reported incidents to be reviewed and managed.	
FR-10	Functional	Traffic Alerts: The system shall generate and display alerts related to congestion, accidents, closures, and hazards.	High	Pass: Alerts are generated and displayed for supported congestion, accident, closure, and hazard events when information is available. Fail: Relevant supported events do not produce/display alerts.	Makes important traffic events visible to users.	
FR-11	Functional	Notifications: The system shall notify relevant users when important traffic events occur in their selected or nearby areas.	Medium	Pass: Relevant users receive a notification for an important event in a selected or nearby area. Fail: Relevant users are not notified or unrelated users are targeted.	Helps users learn about nearby or selected-area traffic events.	

FR-12	Functional	Route Planning: The system shall allow users to enter a source and destination and receive a suitable route.	High	Pass: A user can enter a source and destination and receive a route. Fail: A valid source-destination request returns no route when one is available.	Helps users plan journeys between locations.	
FR-13	Functional	Congestion-Aware Routing: The system shall consider traffic congestion when recommending the least-congested or fastest route, where data is available.	High	Pass: When congestion data is available, route recommendations take it into account. Fail: Available congestion data is ignored in route recommendations.	Improves route recommendations by considering traffic conditions.	
FR-14	Functional	Emergency Priority: The system shall allow authorized emergency vehicles to request priority handling.	High	Pass: An authorized emergency vehicle can submit a priority-handling request. Fail: An unauthorized requester can submit the request or an authorized request cannot be submitted.	Supports priority handling for emergency vehicles.	
FR-15	Functional	Emergency Route: The system shall provide an optimized route for an emergency vehicle based on available traffic information.	High	Pass: An emergency vehicle is provided an optimized route using available traffic information. Fail: Available traffic information is not considered in the route.	Supports emergency travel using traffic-aware routing.	
FR-16	Functional	Parking Availability: The system shall display parking areas and available parking-space information where data is available.	Medium	Pass: Parking areas and available-space information are displayed when data is available. Fail: Available parking information is not displayed.	Helps users identify parking locations and availability.	
FR-17	Functional	Traffic Analytics: The system shall analyze traffic flow, congestion, incidents, and traffic patterns.	Medium	Pass: The system produces analysis covering traffic flow, congestion, incidents, and traffic patterns. Fail: One or more specified areas are not included in the analysis.	Supports understanding of traffic behavior and trends.	
FR-18	Functional	Report Generation: The system shall allow authorized administrators to generate traffic management reports.	Medium	Pass: An authorized administrator can generate a traffic management report. Fail: Report generation is unavailable to an authorized administrator or accessible to an unauthorized user.	Provides summarized information for traffic management.	
FR-19	Functional	Data Management: The system shall store and retrieve users, traffic updates, incidents, alerts, routes, parking data, signal states, and reports.	High	Pass: The system stores and retrieves each listed data category. Fail: A listed category cannot be stored or retrieved as required.	Maintains the information needed by system features.	
FR-20	Functional	Audit Logging: The system shall record important administrative actions such as signal changes, incident updates, and alert creation.	High	Pass: Signal changes, incident updates, alert creation, and other important administrative actions are recorded. Fail: A required administrative action is not logged.	Provides traceability for important administrative operations.	
NFR-01	Nonfunctional	Passwords shall be stored securely, authenticated sessions shall be protected, and role-based access shall prevent unauthorized functions.	High	Pass: Passwords are stored securely, sessions are protected, and role restrictions block unauthorized functions. Fail: Any stated security control is absent or bypassable.	Protects accounts, sessions, and restricted functions.	
NFR-02	Nonfunctional	Normal user requests should receive a response within an acceptable time under the expected project load.	Medium	Pass: Normal user requests respond within the response-time target defined for the project under expected load. Fail: Requests exceed that agreed target under expected load.	Keeps routine interactions responsive. The source does not specify a numeric response-time target.	
NFR-03	Nonfunctional	The system should remain available during normal operating periods, subject to network and service dependencies.	Medium	Pass: The system remains available during normal operating periods when its required network and services are available. Fail: The system is unavailable during normal operation without an identified dependency issue.	Supports access during expected operating periods.	

NFR-04	Nonfunctional	The interface shall be simple, consistent, readable, and suitable for users with basic computer skills.	Medium	Pass: Users with basic computer skills can navigate a consistent, readable interface and complete common tasks. Fail: Interface complexity or readability prevents common tasks.	Makes the system approachable for its intended users.	
NFR-05	Nonfunctional	The system shall validate input and handle errors without corrupting stored data.	High	Pass: Invalid input is validated and errors are handled without corrupting stored data. Fail: Invalid input or an error causes stored data corruption.	Preserves data correctness when errors occur.	
NFR-06	Nonfunctional	The design should permit future addition of more roads, users, sensors, parking areas, and traffic data.	Medium	Pass: The design can be extended to include additional roads, users, sensors, parking areas, and traffic data without redesigning the entire system. Fail: Such additions require fundamental redesign.	Allows the system to grow as requirements expand.	
NFR-07	Nonfunctional	The software shall use modular components so that individual features can be modified with minimal impact on other modules.	Medium	Pass: Features are organized into modules and an individual feature can be changed with minimal impact on other modules. Fail: A local change routinely requires unrelated modules to be changed.	Reduces the effort and risk involved in maintaining the software.	
NFR-08	Nonfunctional	The application should support commonly used modern browsers and standard computing devices.	Medium	Pass: The application operates on the modern browsers and standard computing devices selected for project testing. Fail: It fails to operate on a supported test environment.	Broadens access across common user environments.	
NFR-09	Nonfunctional	The system shall maintain consistent and accurate records for traffic events, incidents, users, and administrative actions.	High	Pass: Records for traffic events, incidents, users, and administrative actions remain consistent and accurate through supported operations. Fail: Records become inconsistent or inaccurate.	Maintains trustworthiness of stored system records.	
NFR-10	Nonfunctional	The system shall collect only necessary user information and protect personal and location-related data.	High	Pass: Only necessary user information is collected and personal/location-related data is protected. Fail: Unnecessary information is collected or protected data is exposed.	Limits unnecessary data collection and protects user information.	
NFR-11	Nonfunctional	The architecture should allow future integration with real traffic sensors, IoT devices, public transport systems, and smart-city infrastructure.	Medium	Pass: The architecture permits future integration with traffic sensors, IoT devices, public transport systems, and smart-city infrastructure. Fail: The architecture prevents such integrations without major redesign.	Supports future integrations beyond the prototype.	
NFR-12	Nonfunctional	Critical administrative operations shall be traceable through logs containing relevant action and time information.	High	Pass: Logs record critical administrative actions with relevant action and time information. Fail: A critical operation cannot be traced from the logs.	Supports accountability and investigation of administrative operations.	